

Project 1: Flask App CI/CD with Jenkins Master-Agent Setup + PostgreSQL

Goal:

Deploy a Flask app to multiple EC2s using Jenkins and Ansible roles. Store build artifacts and use a PostgreSQL database.

Requirements:

- Jenkins master: Amazon Linux 2
- Jenkins agent: Ubuntu 22.04
- App EC2s: Centos

Ansible roles:

- Install Python + Flask
- Install and configure PostgreSQL
- Copy app
- Connect app to PostgreSQL (via environment vars or config)

Artifact:

- Jenkins stores app package (.tar.gz) in an internal artifact repo (e.g., local or S3)
- Versioned deploy via Jenkins parameters

Pipeline:

- Jenkins pulls the code from GitHub
- Builds and packages the Flask app
- Archives the artifact

- Triggers Ansible playbook on Jenkins agent
- Ansible deploys the app and connects to PostgreSQL
- App is accessible publicly via browser (port 5000)

Expected Output:

- Web page displays: "I am almost a DevOps Engineer!"